

1. Designing a Standalone PV Energy System

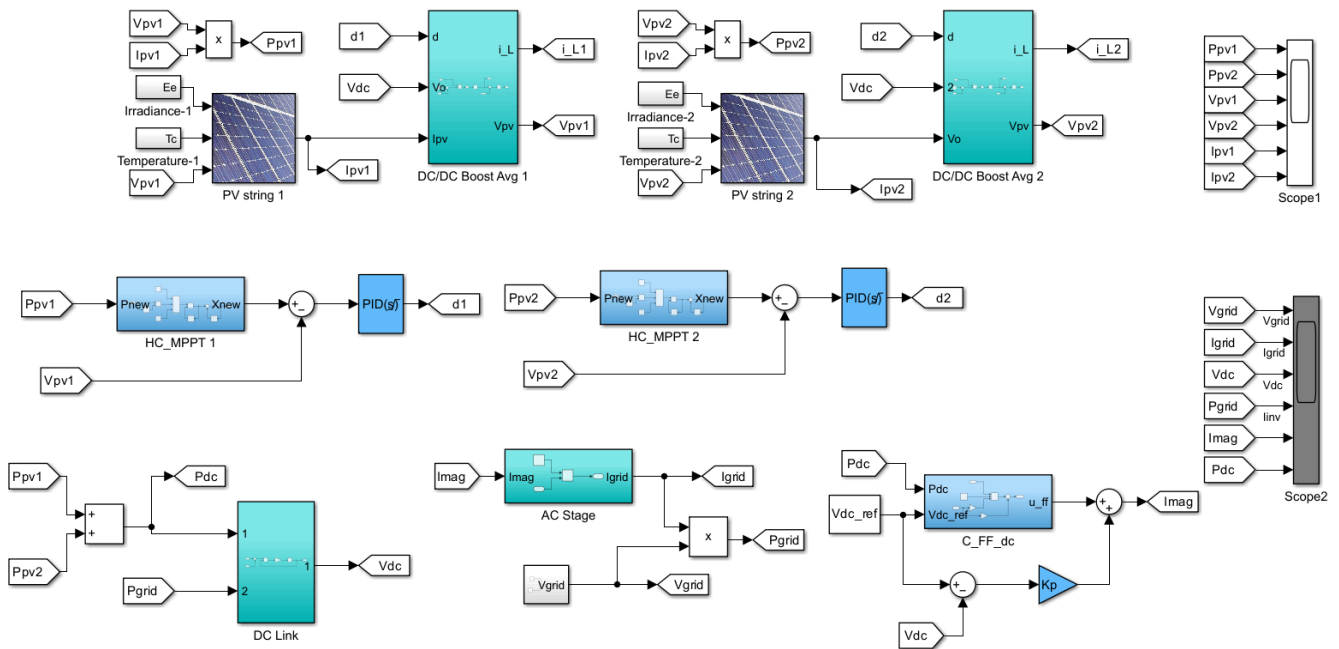


Figure 1.1 overall double-stage PV system

Figure 1.1 represents the full dual stage PV system from the photovoltaic power generation to the grid.

Simulation of Grid-Connected Multi-String Inverters

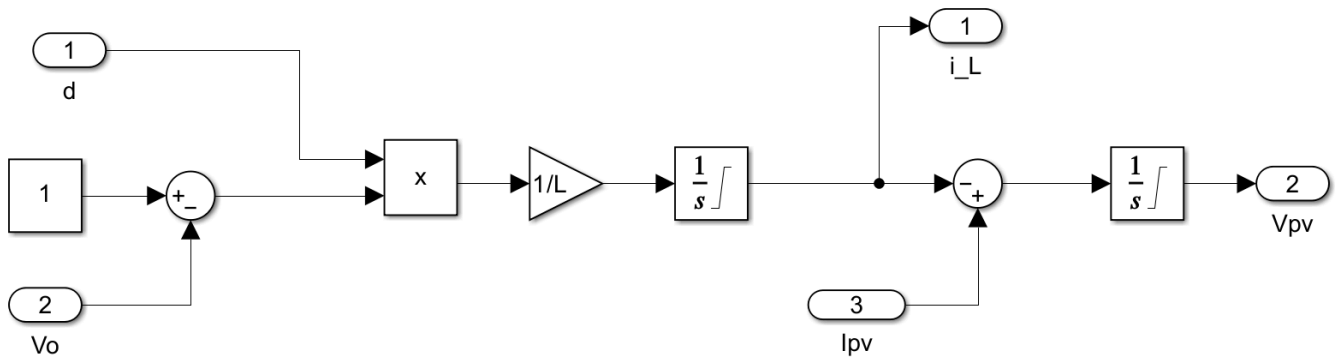


Figure 1.2 DC/DC boost converter model

Figure 1.2 represents the DC to DC boost converter that takes in a duty cycle and output voltage of the PV generator and outputs a boosted or higher voltage.

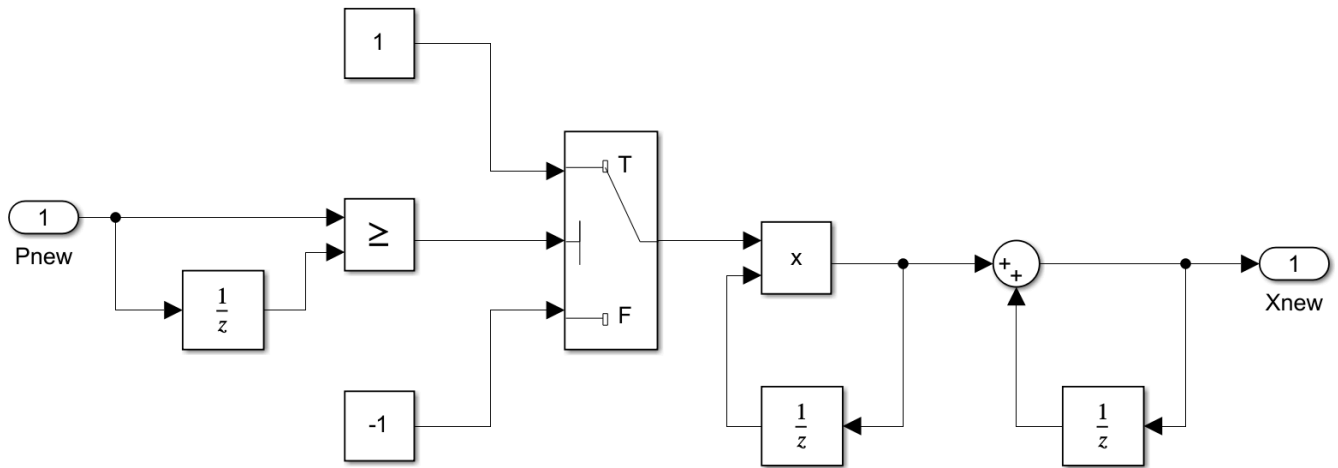


Figure 1.3 HC_MPPT model

Figure 1.3 represents the Hill Climbing Maximum Power Point Tracking algorithm used within the system. Its job is to iteratively identify the maximum power point by continuously making changes until the power no longer increases.

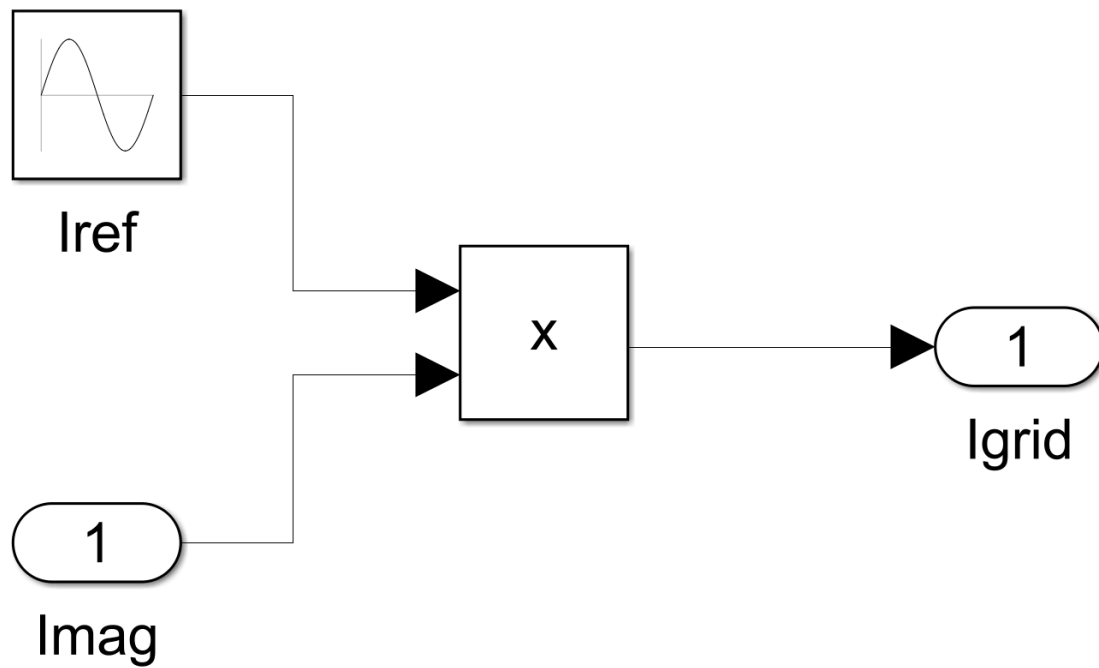


Figure 1.4 AC stage model

Figure 1.4 represents the AC subsystem modeled in our PV system. It simulates the current output on the grid by multiplying the magnitude of the input current with a sine wave

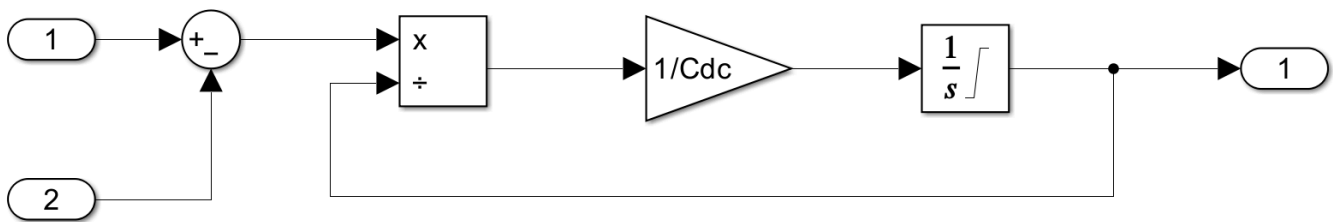


Figure 1.5 DC Link

This model represents the DC link capacitor of the system that supports the DC/DC converter and PV array. Its job is to stabilize the link by providing power when the converter needs it and acting as a sink when there is excess energy at the converter.

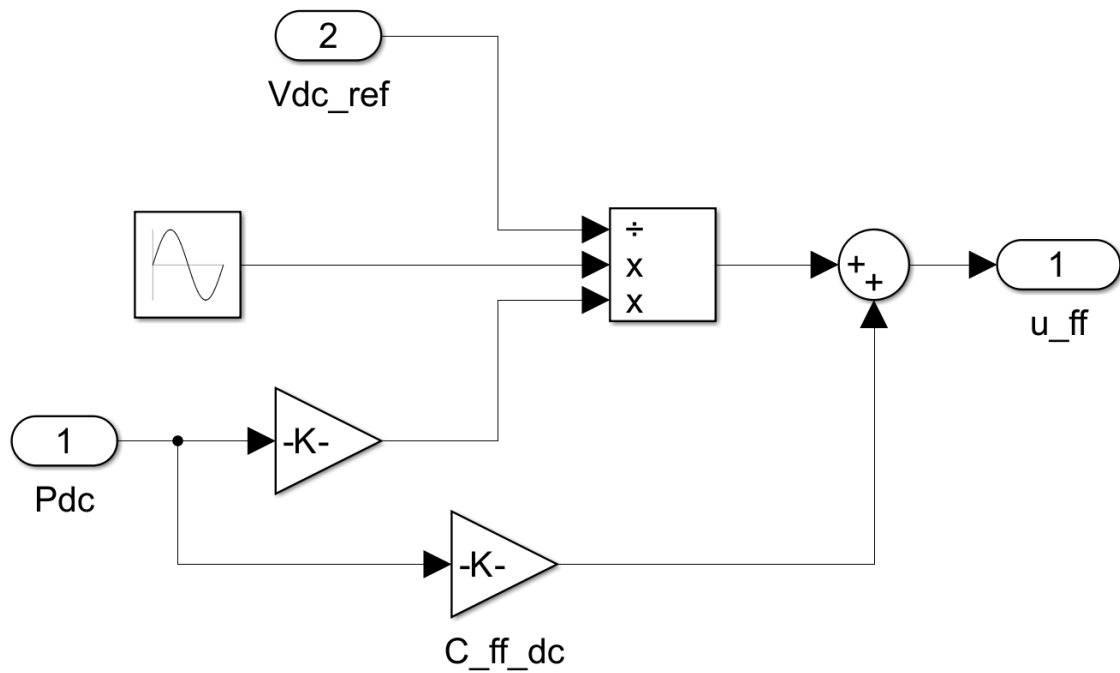


Figure 1.6: Feed Forward Coefficient

Figure 1.6 represents the feed forward coefficient subsystem that scales the DC power down to generate the feed-forward control signal that affect the inverter behavior.

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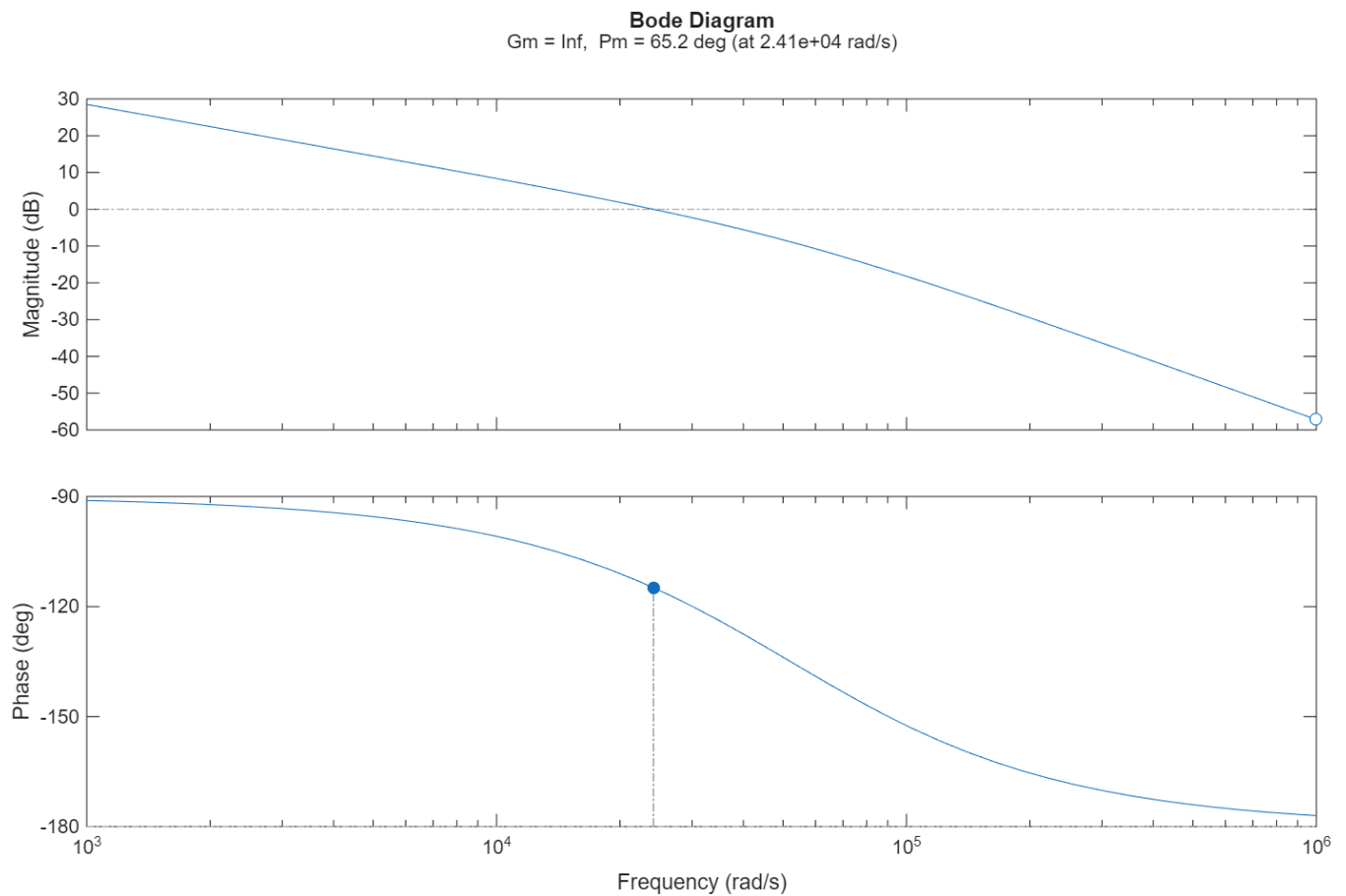


Figure 1.7: Transfer Function Bode Plot

Figure 1.7 simultaneously displays the magnitude and phase plots for the transfer function of the control loop in the PV system. We can observe from this figure that the phase margin is at approximately 65 degrees, with an infinite gain margin.

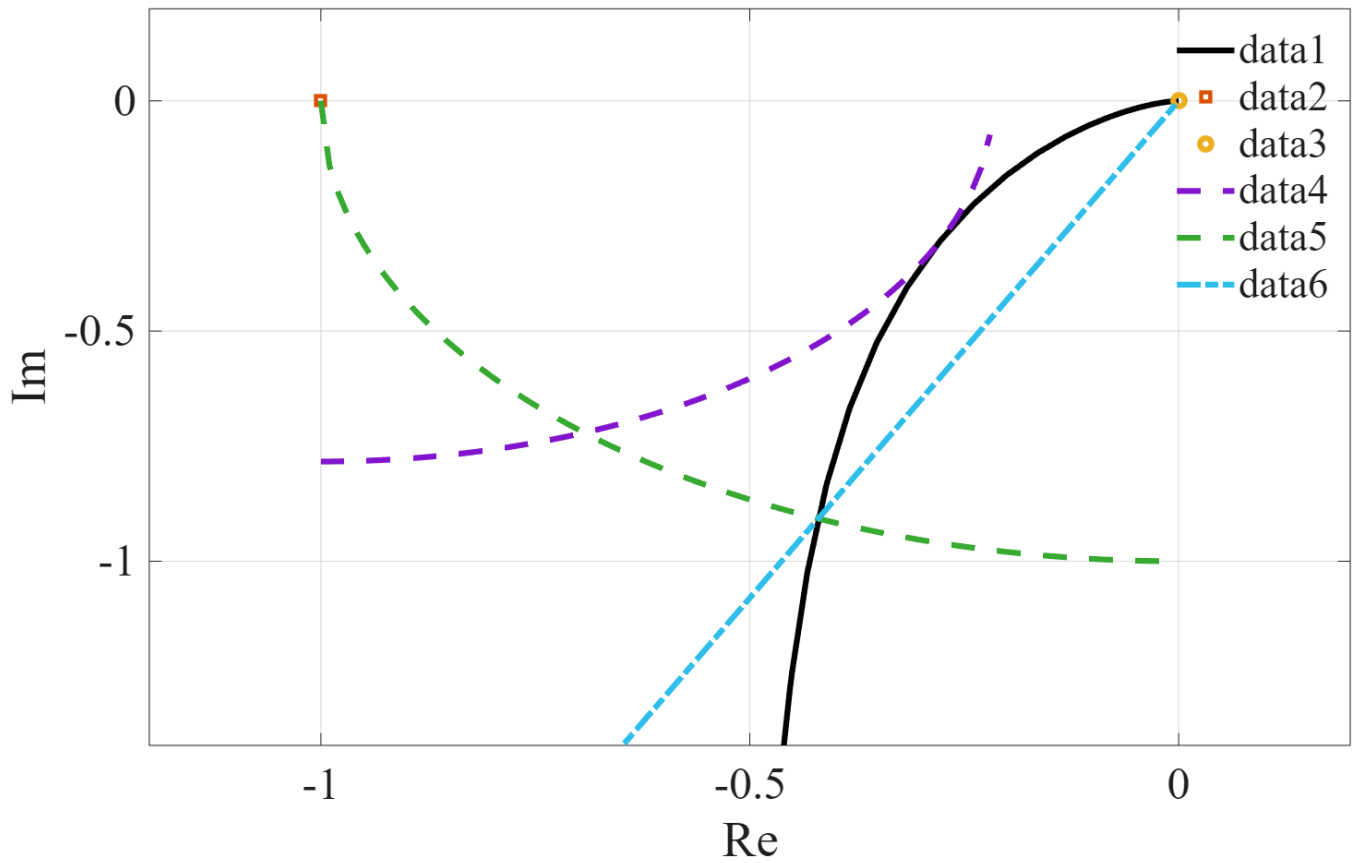


Figure 1.8: Nyquist Plot

Figure 1.8 displays the Nyquist plot of the control loop for the PV system. It represents the frequency response on the complex plane, with the horizontal axis being the real component. This type of plot is used to determine stability of control loops. In this case, since the plot does not encircle the critical point $(-1, 0)$, we can say the control loop is stable.

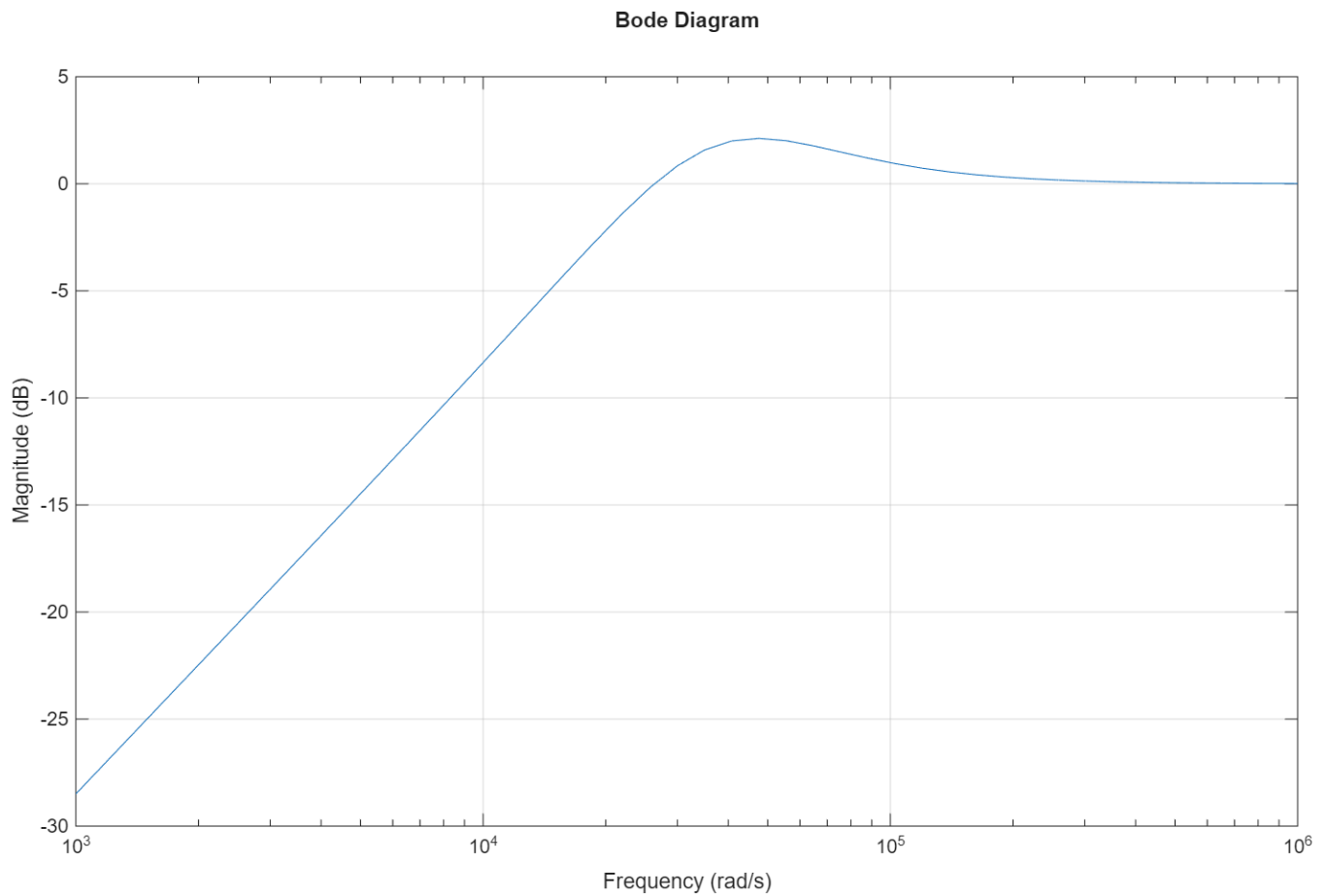


Figure 1.9: Feed Forward Frequency Response

Figure 1.9 showcases the frequency response of the feed forward subsystem. As can be seen in the figure, it shows that lower frequencies are attenuated while higher frequencies are allowed through at almost unity gain.

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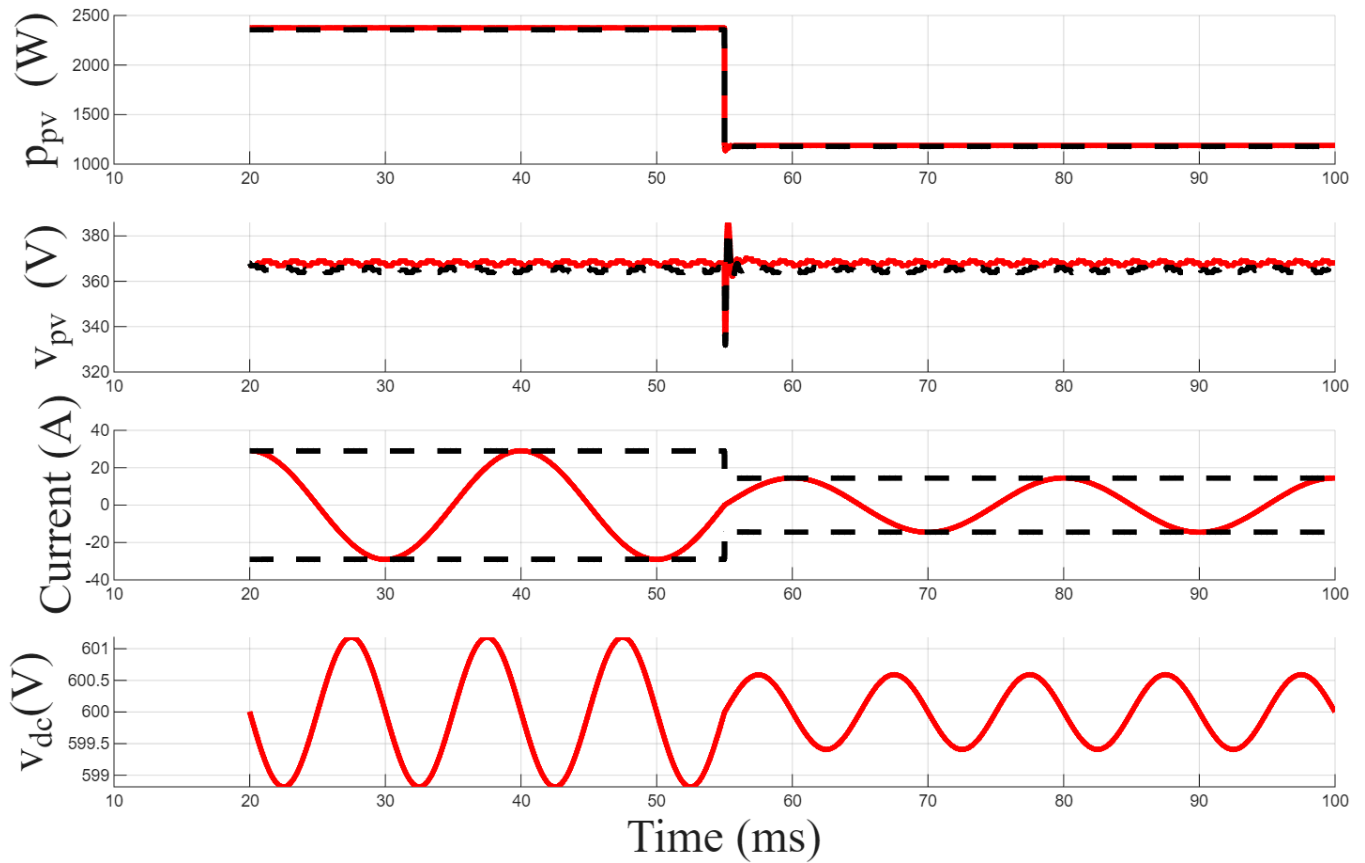


Figure 1.10: Parameter Step Response

Figure 1.10 showcases the step response of system components like the voltage current and power of both the panels and the converter. It demonstrates how the system reacts to a sudden change in solar input conditions, like irradiance. Starting from the first plot, we can see that the power of the system drops significantly at around the 55ms mark. This is due to the lower solar input conditions. Similarly, the DC voltage and current also show lower amplitudes. However, the array voltage does not change much aside from a dip/spike at the time of the step. This shows how the MPPT algorithm is maintaining the operating point of the array.

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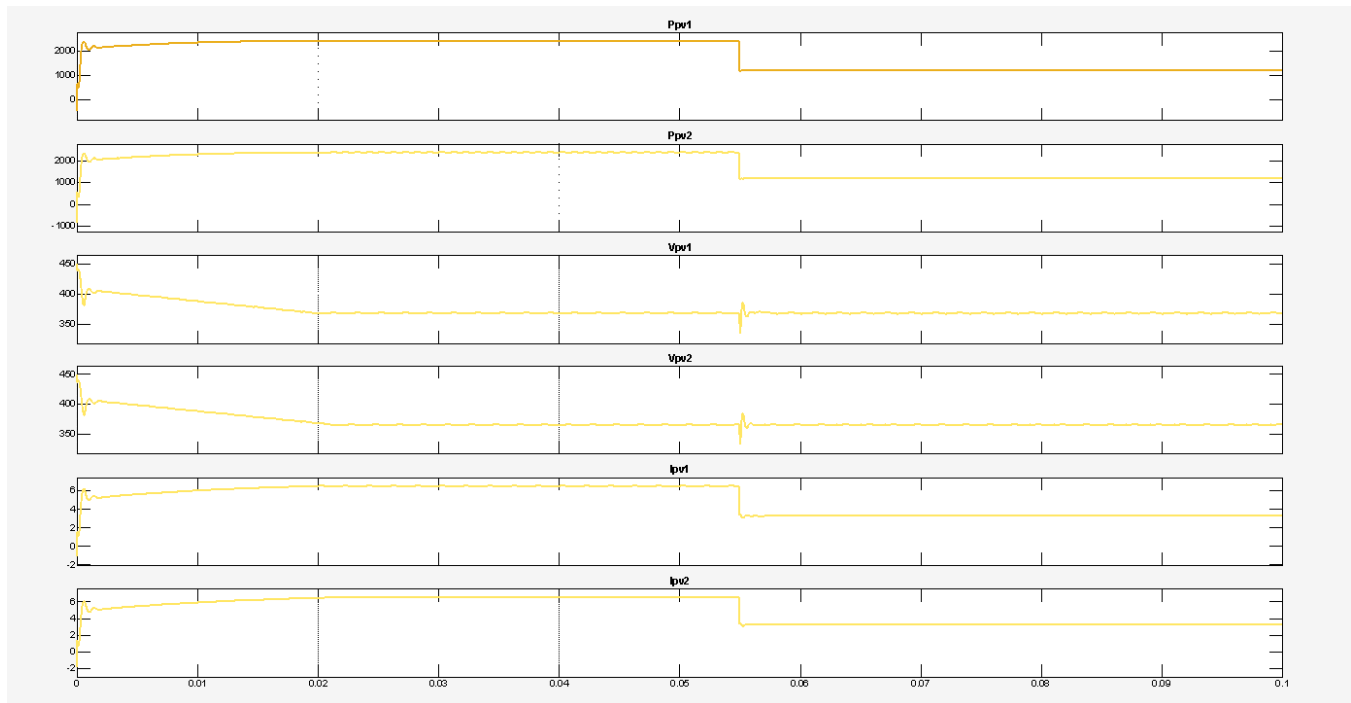


Figure 1.11

Figure 1.11 shows similar step responses as figure 1.10. A noticeable dip is observed at about 55ms because of a sudden change to solar conditions like irradiance. This change lowers parameters like power and currents. Interestingly, and like figure 1.10, the array voltages are maintained relatively stable after the step, demonstrating the maintenance of the operating point provided by the MPPT algorithm.

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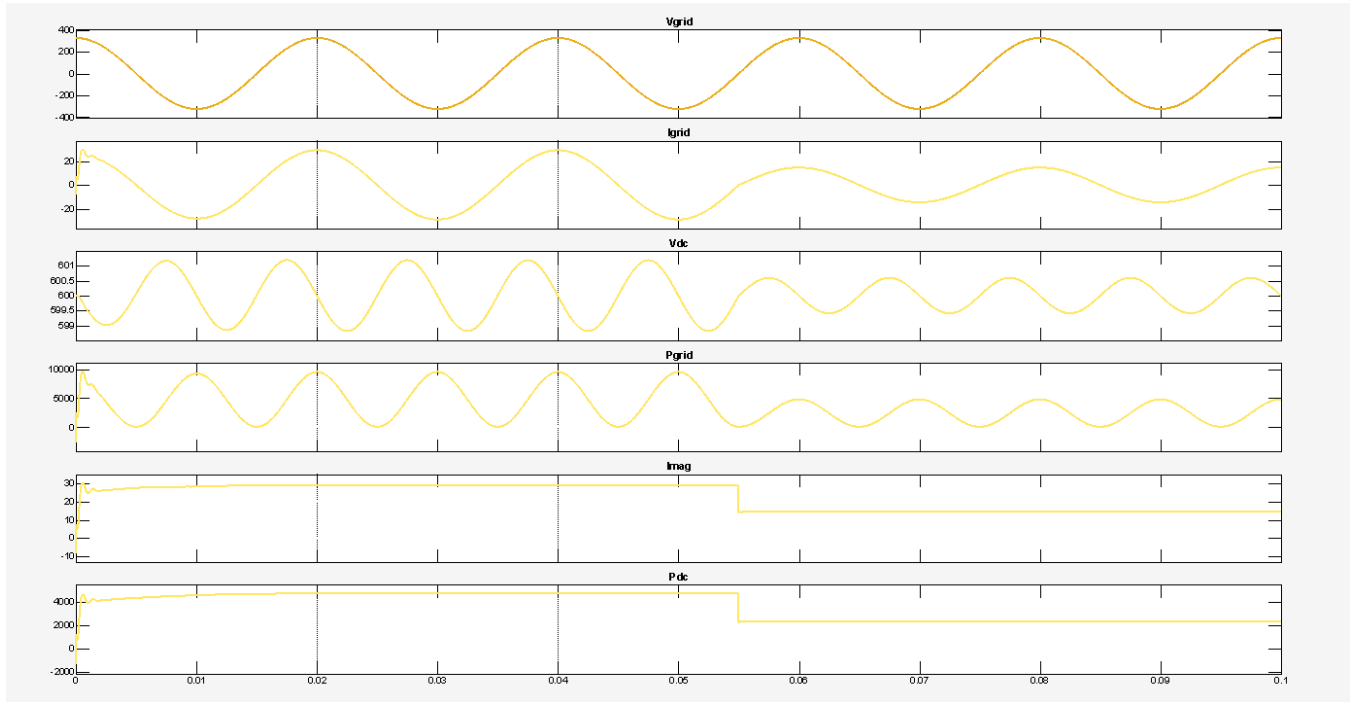


Figure 1.12 Grid-side step response

Figure 1.12 shows the step response of the grid side of the system in response to a sudden and significant change to irradiance conditions. As can be seen in the figure, both the power and current drop once again after the step. The voltage and power on both sides of the inverter show a significant drop after the change in irradiance. This is also true with the current, and is a result of lower energy generation due to lower irradiance.